Three-Model Pipeline: SSM -> CTM -> CDM

Discrete State Space · 1st-Order Traffic Flow · PDE Dispersion

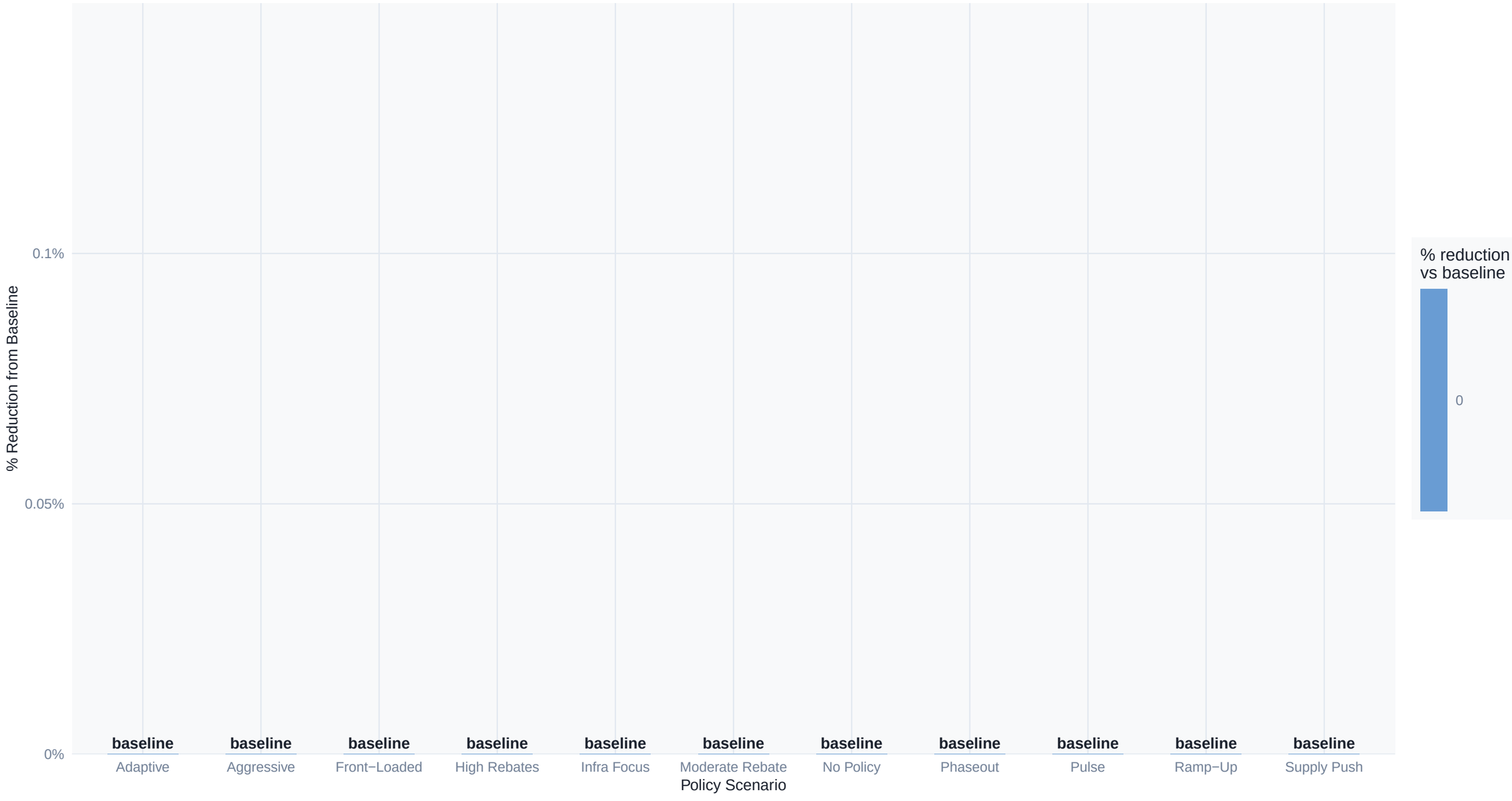
Each model feeds its key transfer variable into the next stage. SSM sets fleet composition, CTM converts fleet to corridor-level flows, CDM solves the advection-diffusion PDE to yield spatially-resolved PM2.5.

PM2.5 AV Scenarios − Year 0

Policies: No Policy | High Rebates | Infra Focus | Aggressive | Moderate Rebate | Supply Push | Phaseout | Ramp−Up | Pulse | Adaptive | Front−Loaded

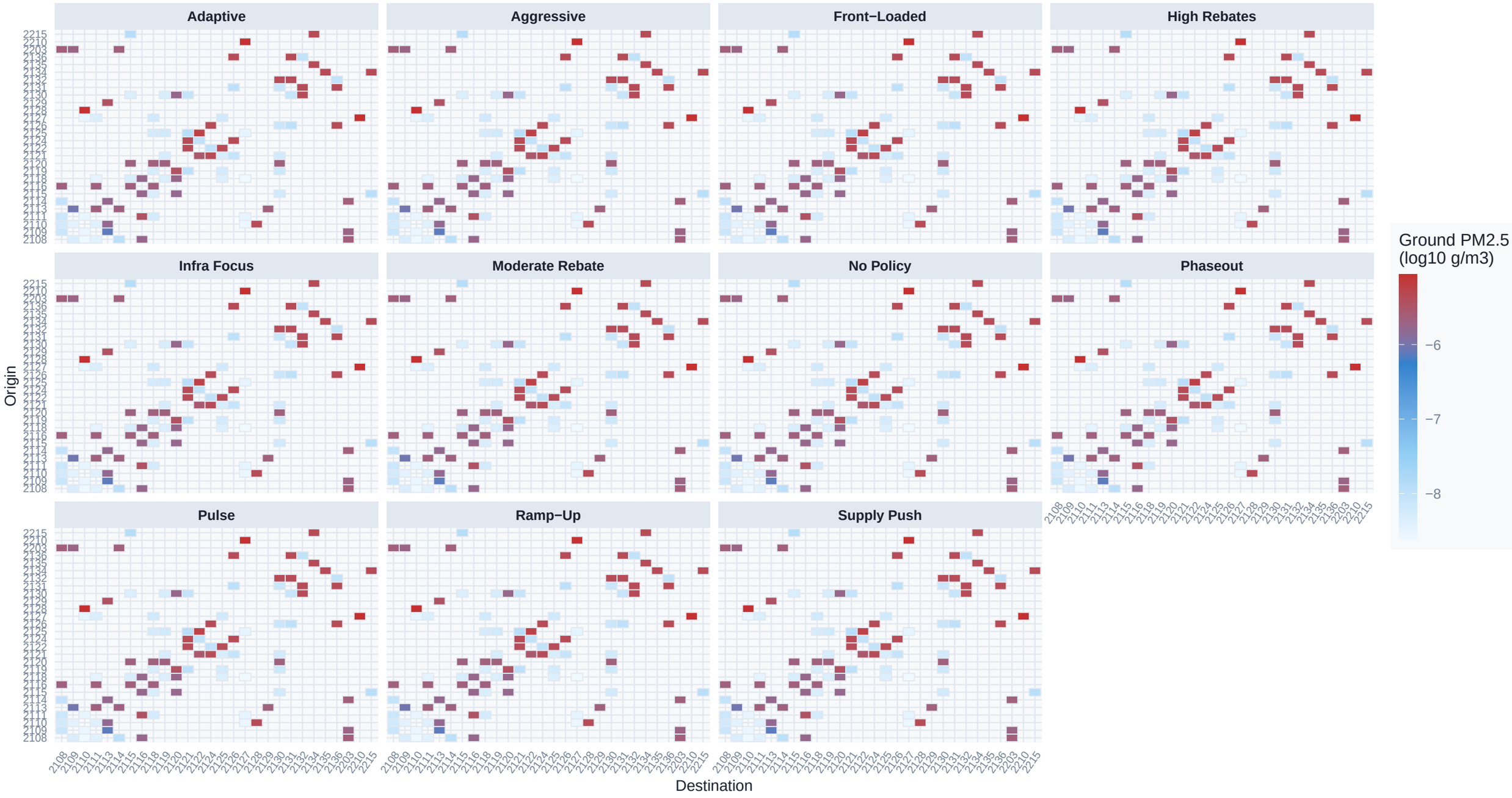
Total Network Emission Rate by Policy Scenario

Sum of E_eff across all active flow corridors; % reduction vs baseline scenario



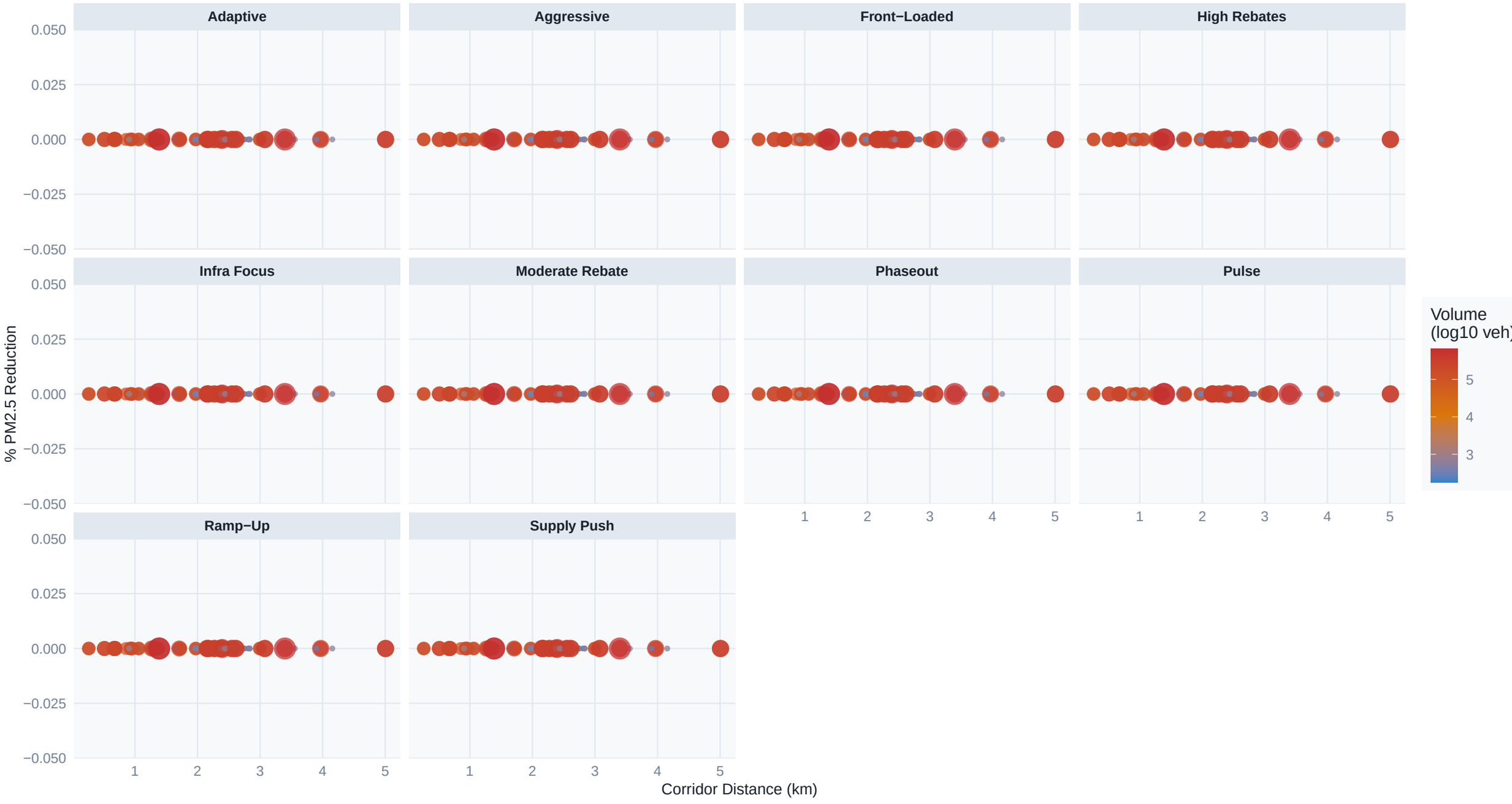
Ground−Level Peak PM2.5 by Flow Pair

Each cell = one origin−destination corridor, color = peak ground concentration



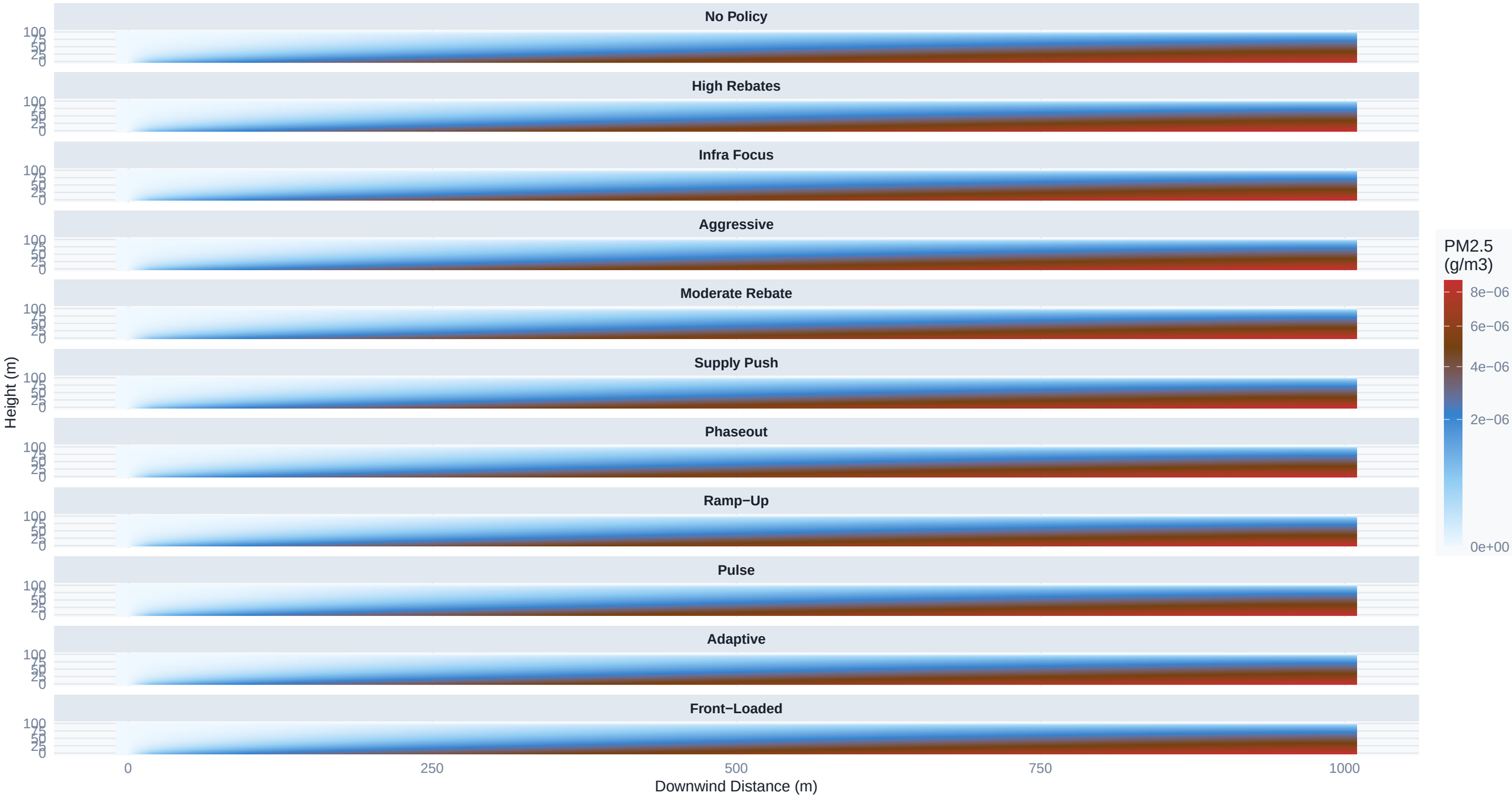
PM2.5 Reduction by Corridor vs Baseline

Each point = one flow pair; x = distance, y = % reduction in ground−level PM2.5



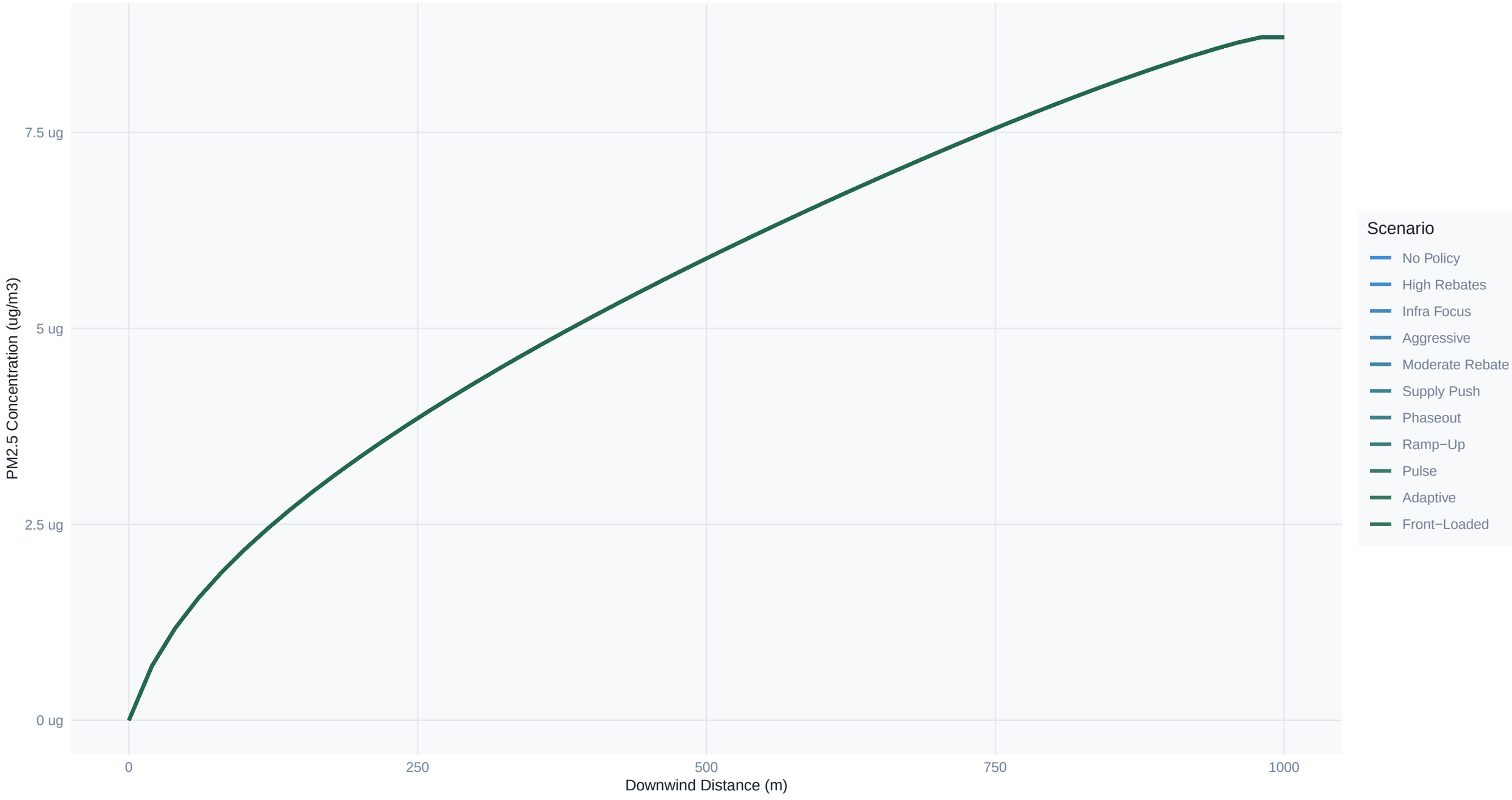
PM2.5 Plume − Busiest Corridor (2127 − 2210)

659,690 vehicles/month, 1.4 km



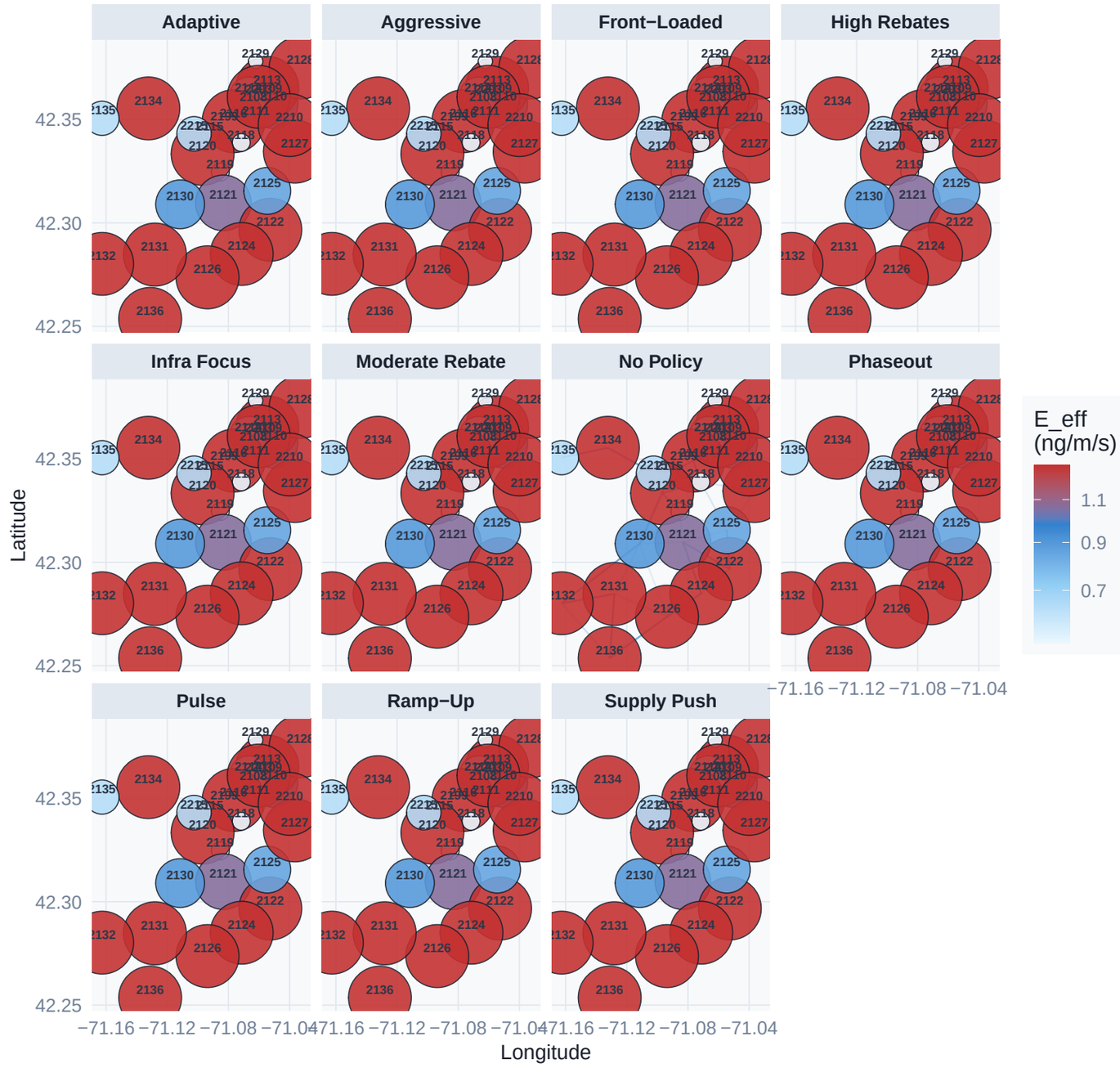
Ground−Level PM2.5 Profile − 2127 − 2210

Concentration along downwind axis at z = 0 (street level), converted to ug/m3



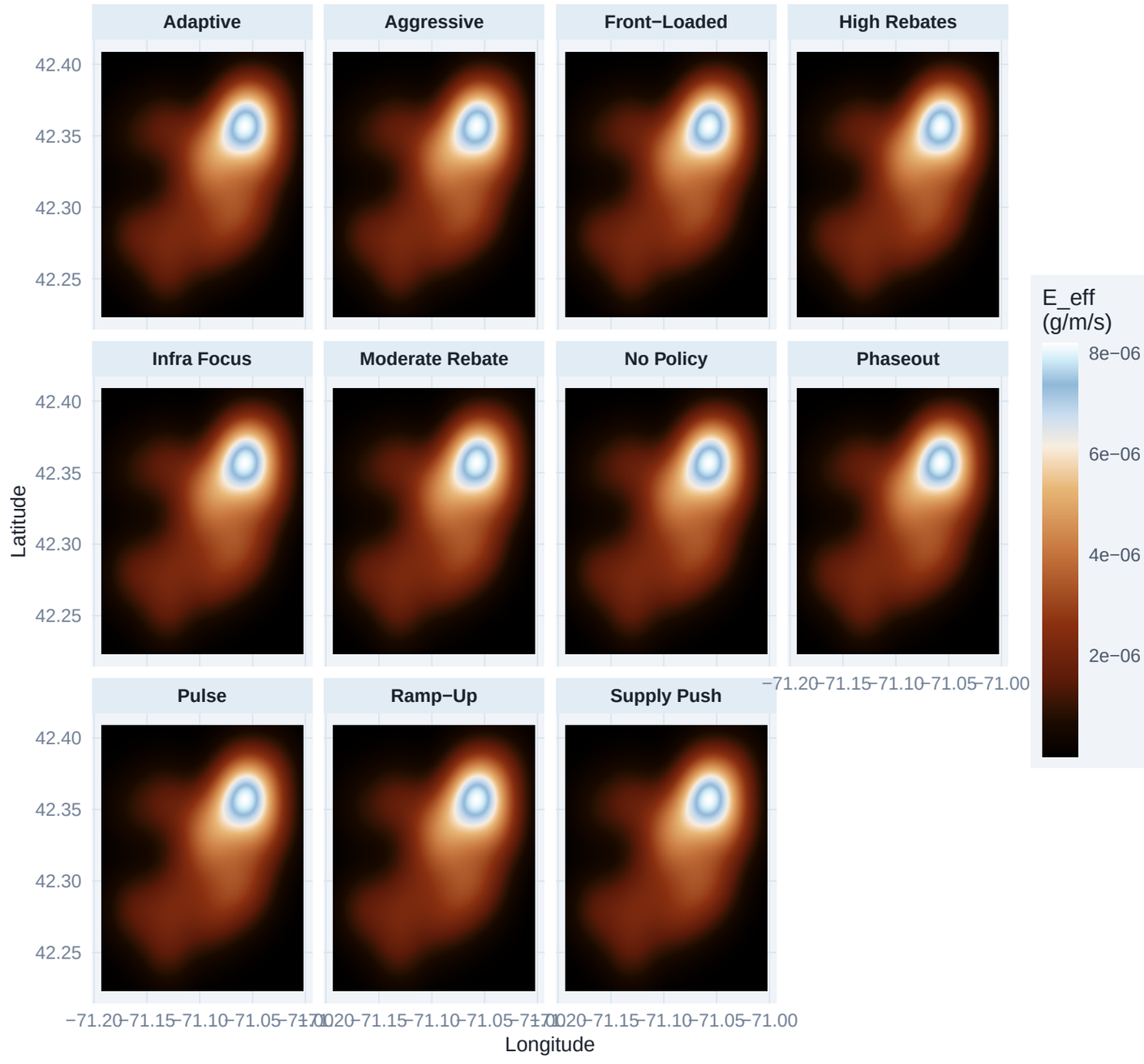
Spatial Distribution of PM2.5 Emission Burden

Node size = outbound vehicle volume, color = total emission rate



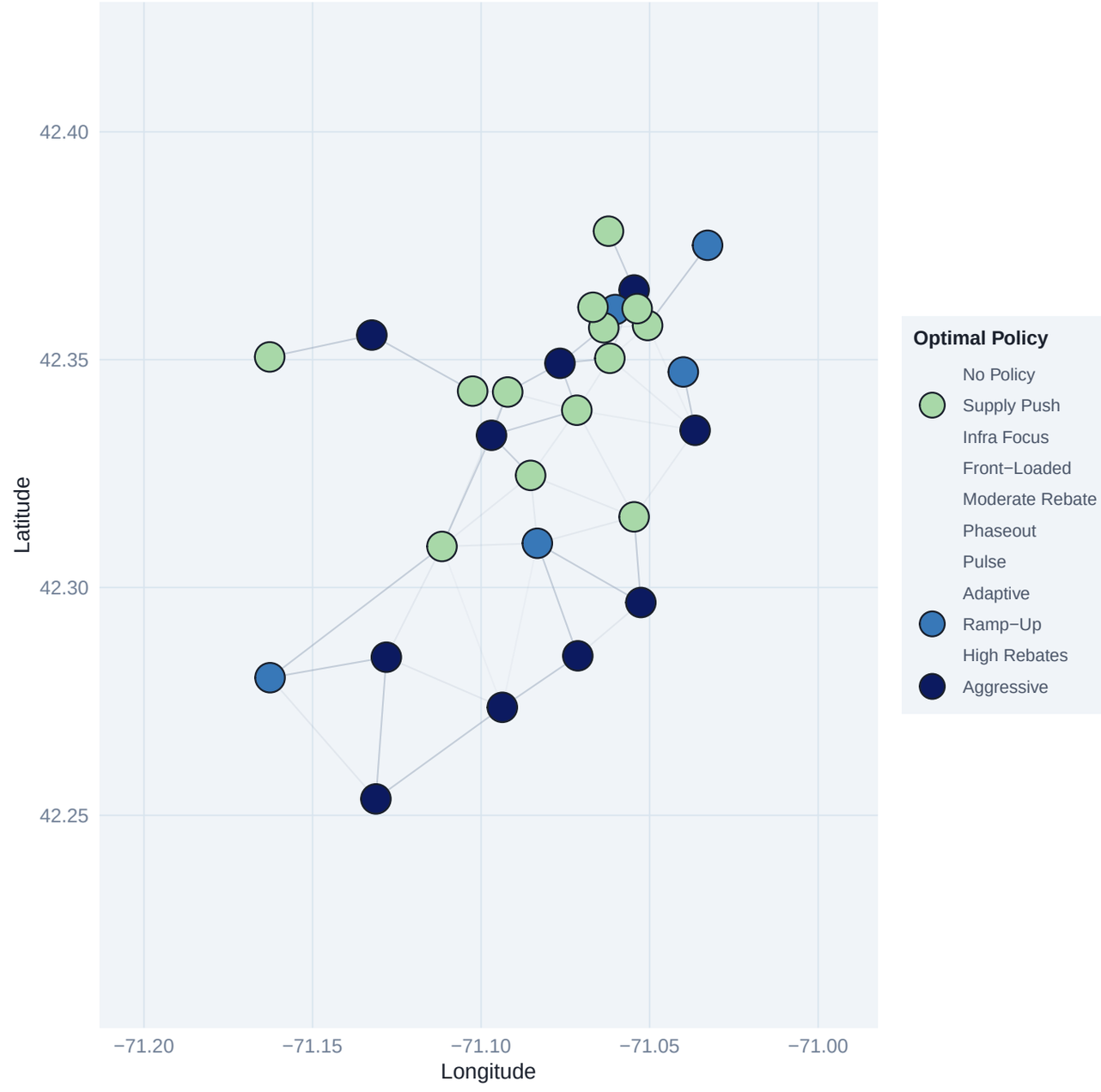
Spatial Distribution of PM2.5 Emission Burden (Absolute)

Gaussian emission surface per zone, warmer color = more pollution



Budget−Optimal Policy per Node ($5.0B cap)

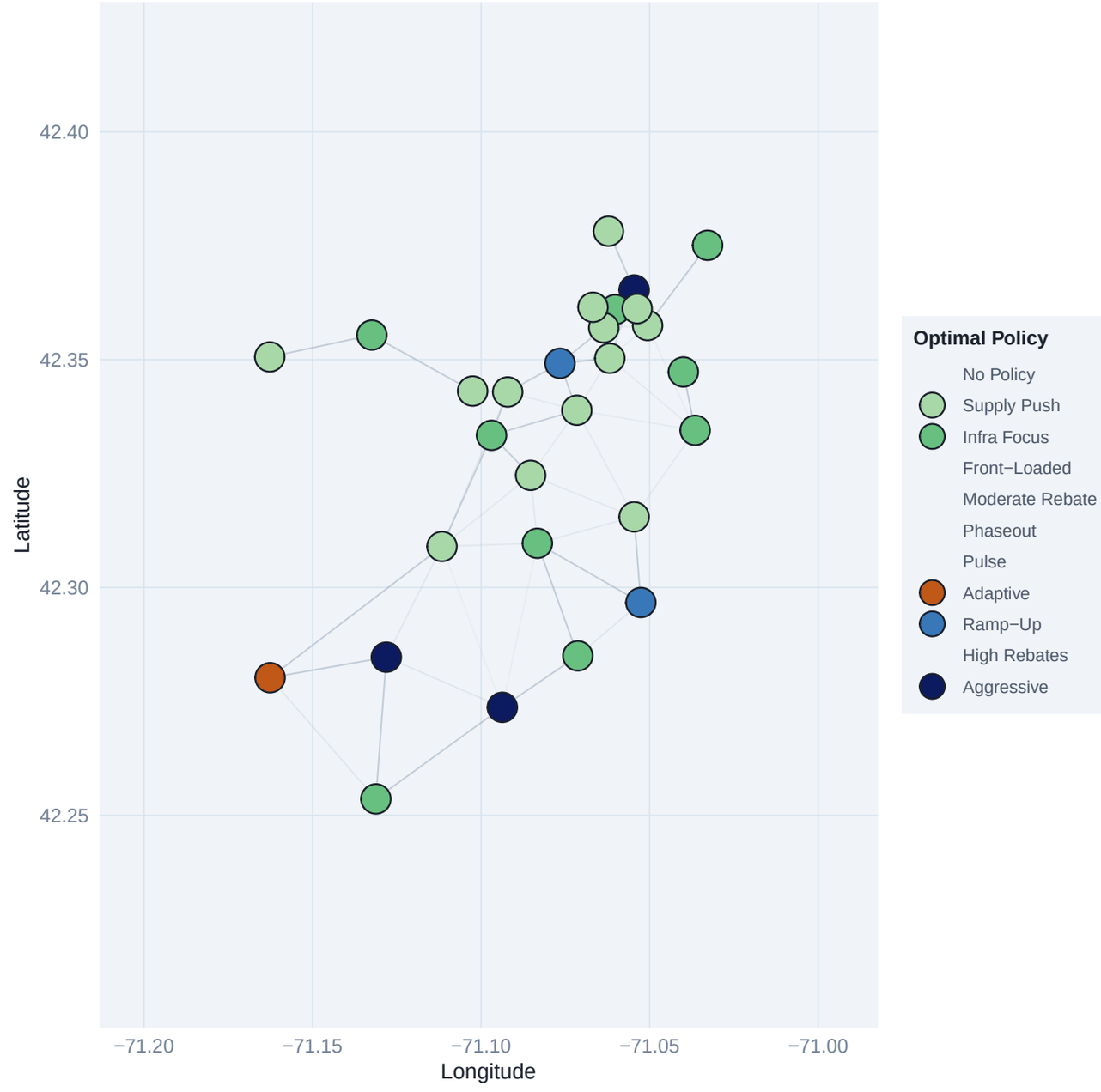
Budget: $5.0B | Spent: $4.97B (99.4%) | Remaining: $0.03B | Policy intensity matched to node PM2.5 burden



Green = low−cost · Blue = moderate · Navy = high−investment · Grey = no policy (budget exhausted)

Budget−Optimal Policy per Node ($2.5B cap)

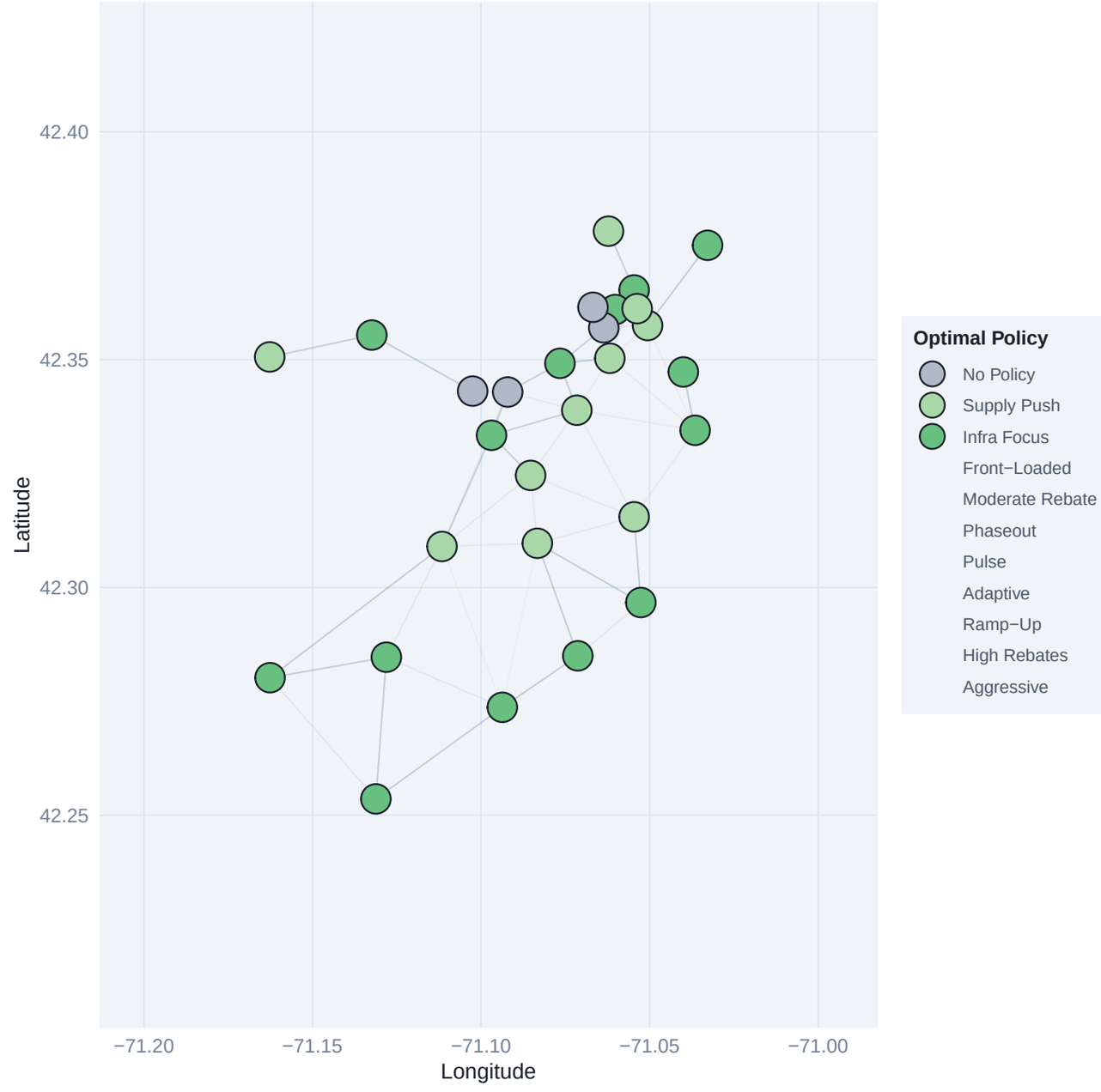
Budget: $2.5B | Spent: $2.47B (98.7%) | Remaining: $0.03B | Policy intensity matched to node PM2.5 burden



Green = low−cost · Blue = moderate · Navy = high−investment · Grey = no policy (budget exhausted)

Budget−Optimal Policy per Node ($1.0B cap)

Budget: $1.0B | Spent: $0.97B (97.1%) | Remaining: $0.03B | Policy intensity matched to node PM2.5 burden



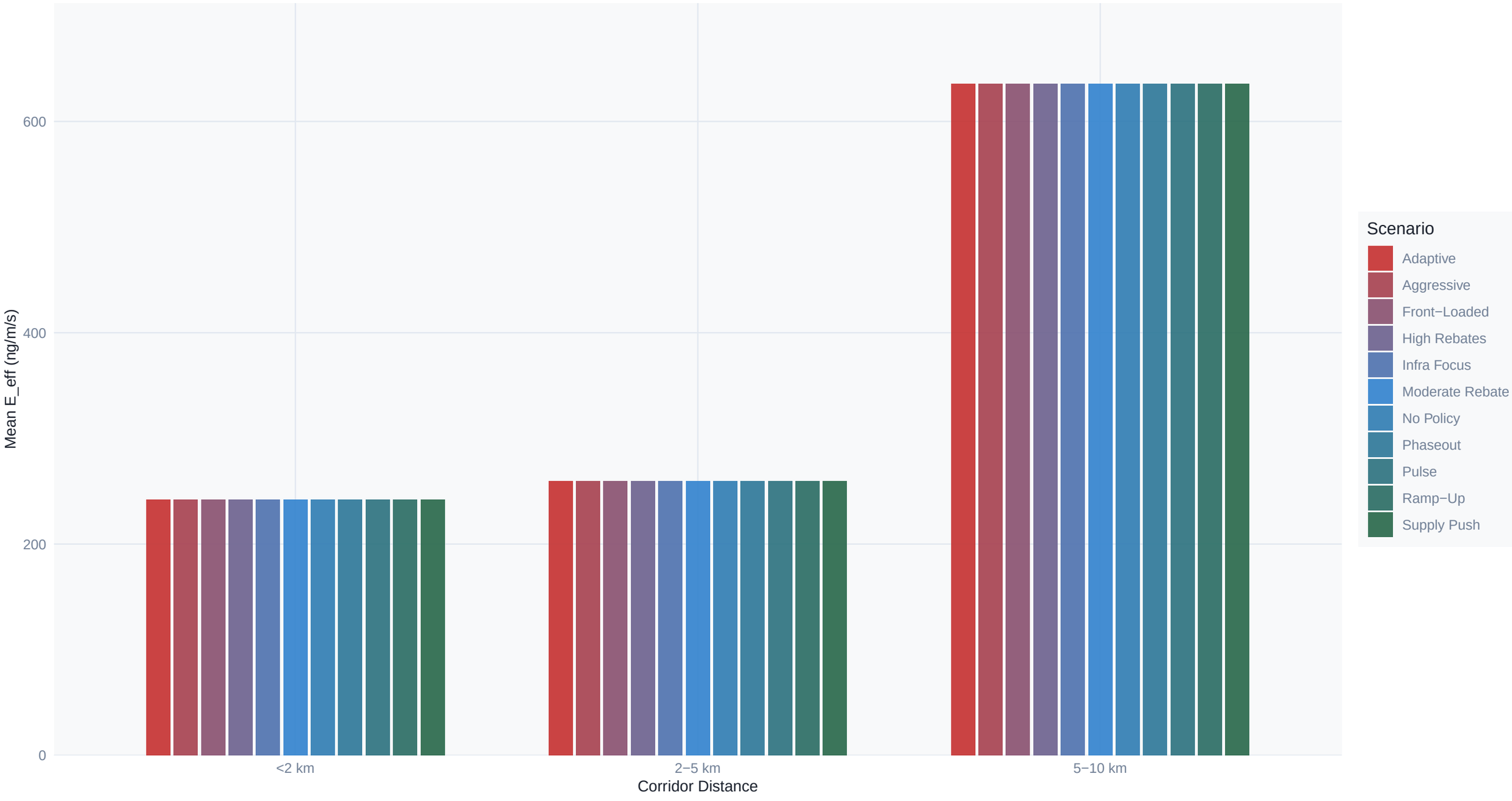
Green = low−cost · Blue = moderate · Navy = high−investment · Grey = no policy (budget exhausted)

Spatial PM2.5 Emission Burden: Deviation from Cross−Scenario Mean

Not shown: all scenarios have identical AV fractions at this year − no variation to display.

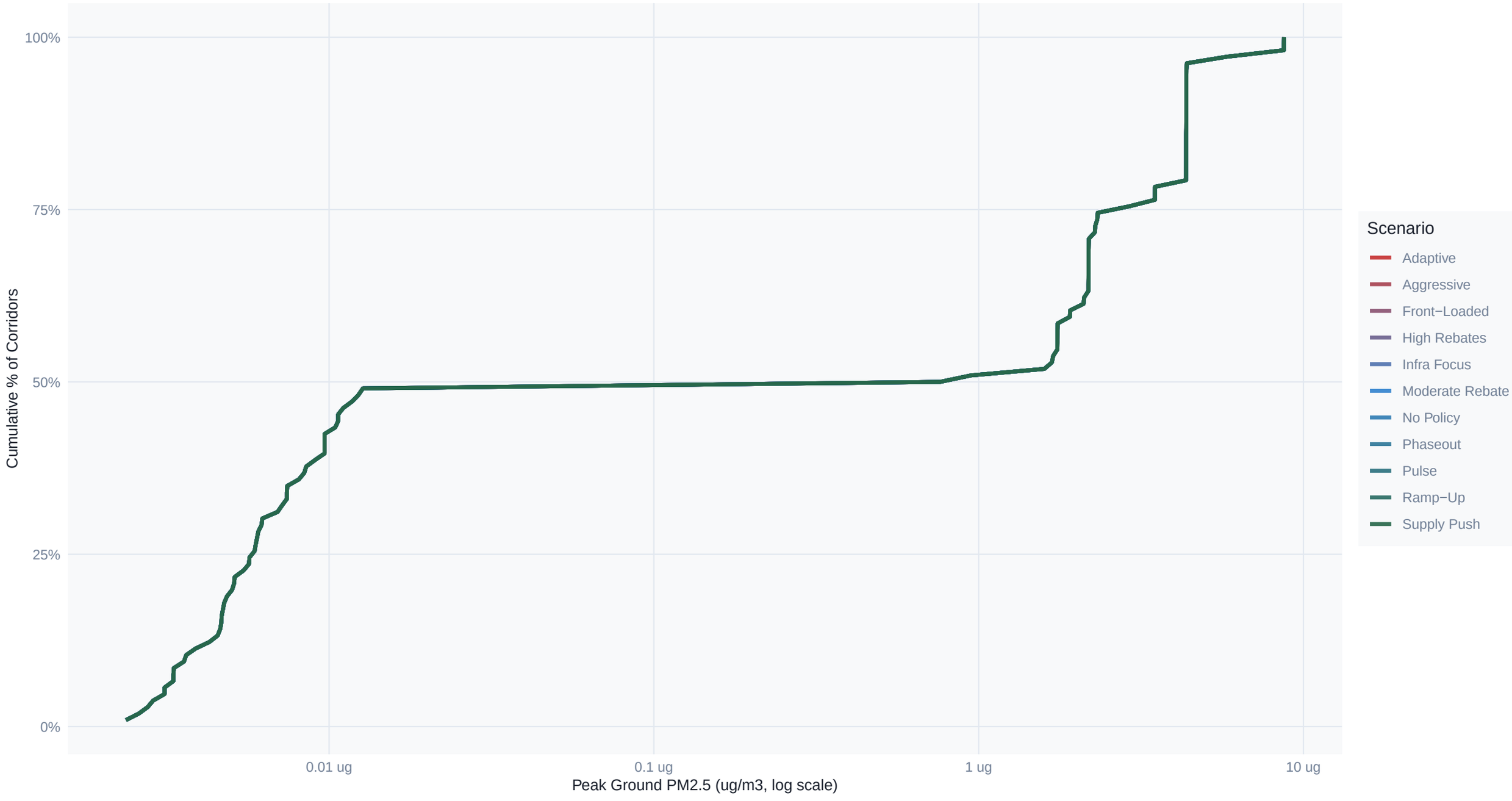
Mean Emission Rate by Corridor Distance Band

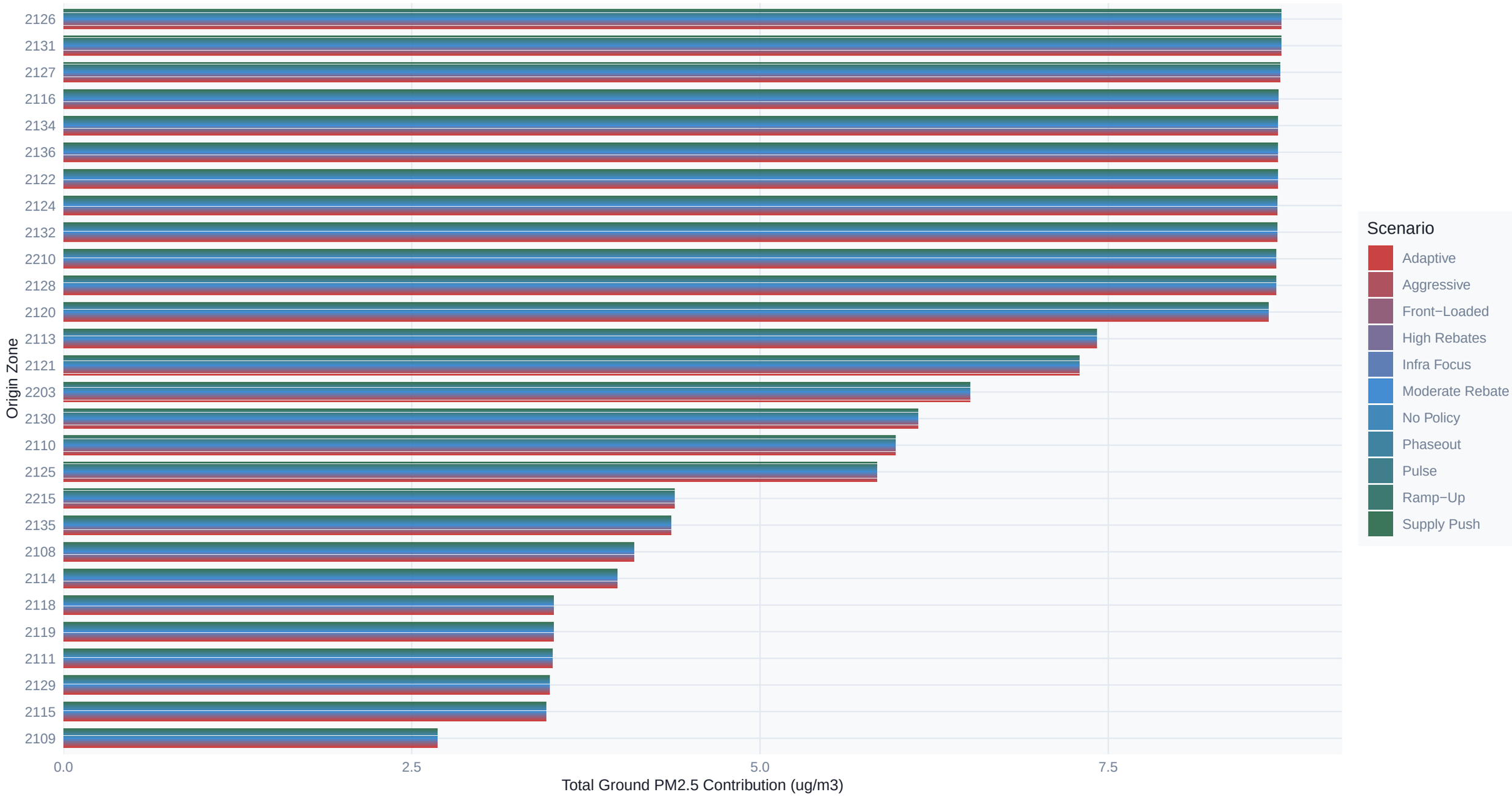
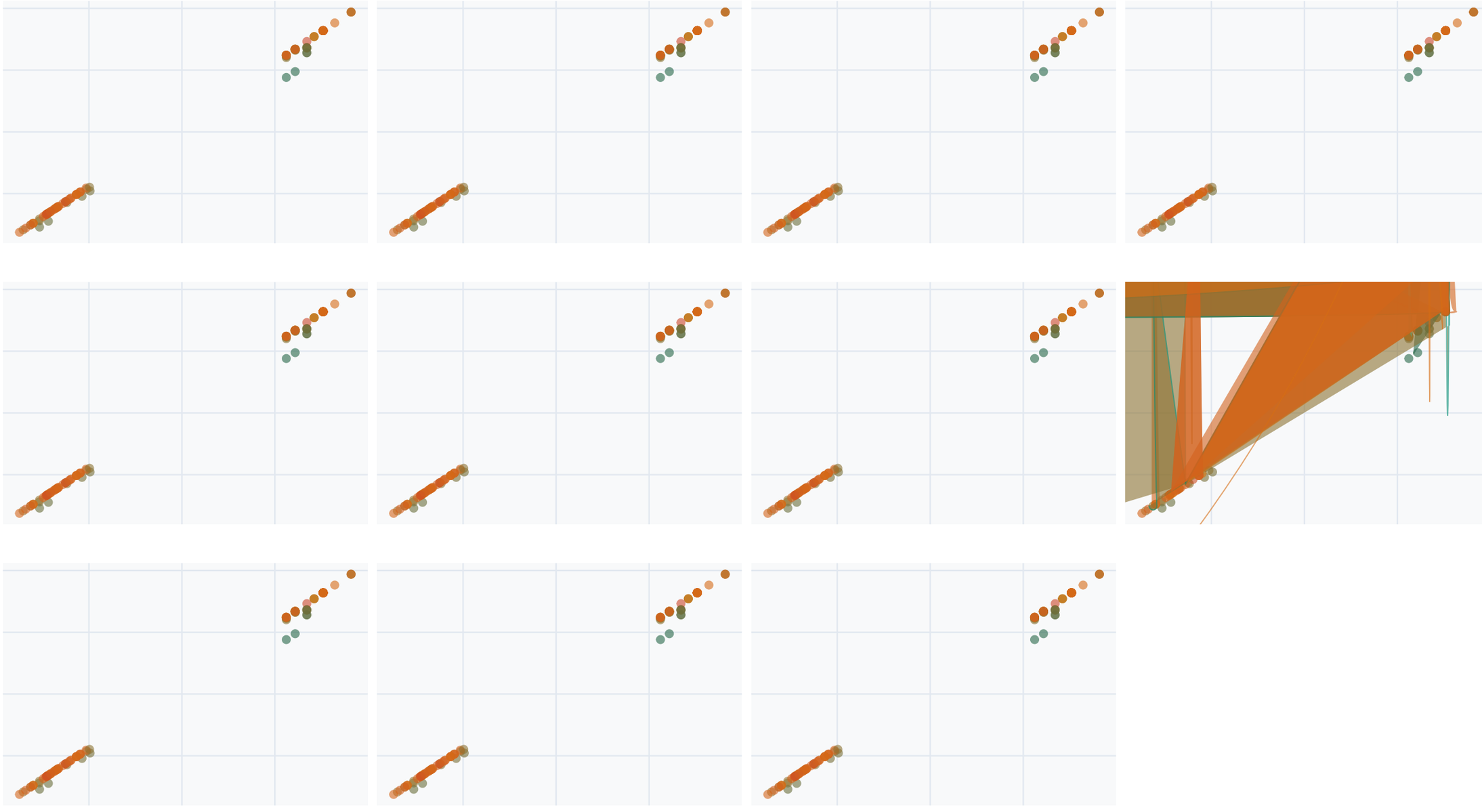
Average E_eff per corridor length category



Cumulative Distribution of Peak Ground−Level PM2.5

Each point = one corridor, rightward shift = worse air quality



PM2.5 Burden by Origin Zone

Sum of peak ground concentrations across all outbound corridors per zone